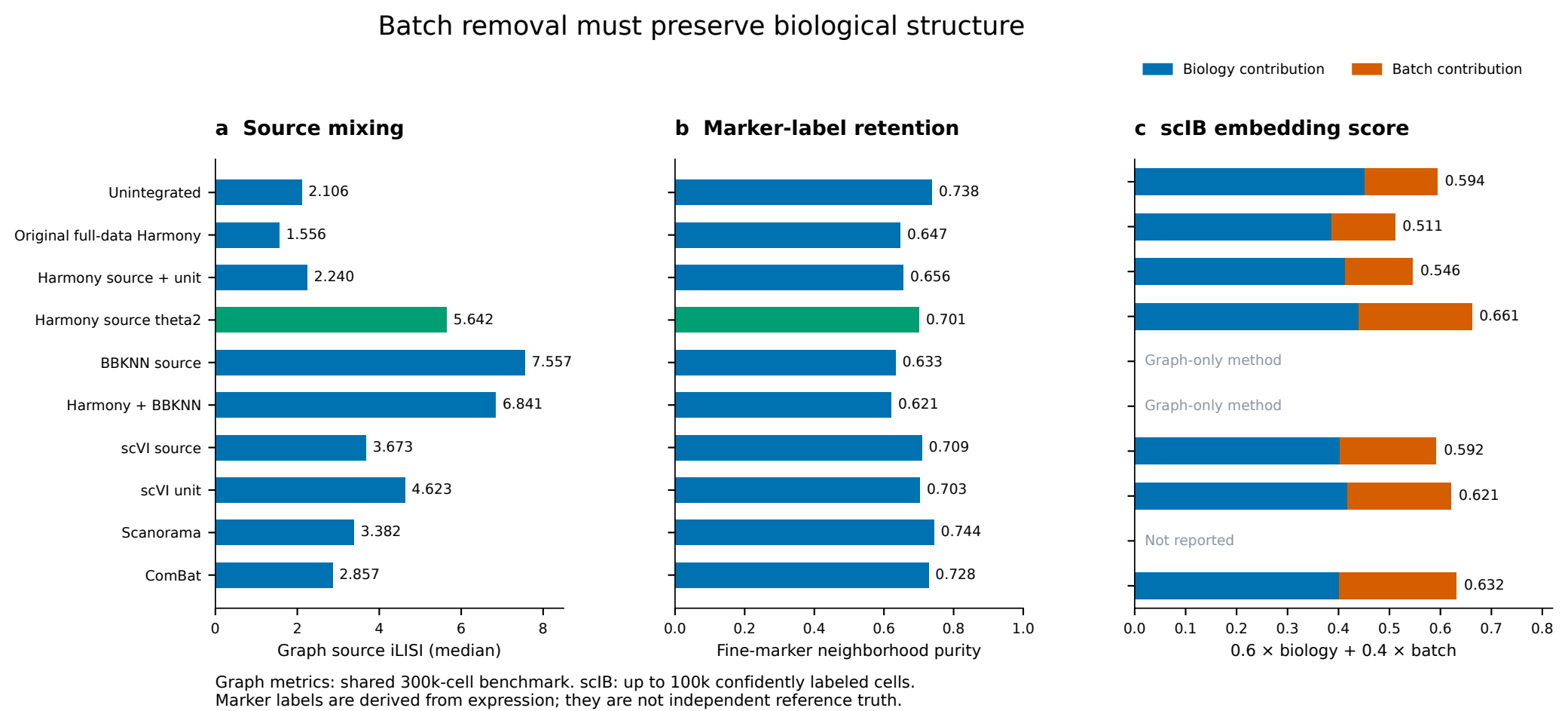
Batch removal must preserve biological structure
Biology contribution
Batch contribution
a Source mixing
Unintegrated
Original full-data Harmony
Harmony source + unit
Harmony source theta2
BBKNN source
Harmony + BBKNN
scVI source
scVI unit
Scanorama
ComBat
2.106
1.556
2.240
5.642
7.557
6.841
3.673
4.623
3.382
2.857
0
2
4
6
8
Graph source iLISI (median)
b Marker-label retention
0.738
0.647
0.656
0.701
0.633
0.621
0.709
0.703
0.744
0.728
0.0
0.2
0.4
0.6
0.8
1.0
Fine-marker neighborhood purity
c scIB embedding score
0.594
0.511
0.546
0.661
Graph-only method
Graph-only method
0.592
0.621
Not reported
0.632
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.6 × biology + 0.4 × batch
Graph metrics: shared 300k-cell benchmark. scIB: up to 100k confidently labeled cells.
Marker labels are derived from expression; they are not independent reference truth.